The Eyes of Texas are Upon OB/GYNs: Physician Migration and Crowdsourced Enforcement of Abortion Regulations

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Dissertation Proposal

How does the U.S. Constitution guarantee rights?

Ex: *Brown v. Board of Education* (school segregation)

- The federal constitution guarantees a right, but a state passes a policy infringing on that constitutional right
- Someone sues the state, specifically the enforcer of the offending policy
- A judge decides if the state policy infringes on the constitutional right
- But, what if there's no gov't enforcer of the offending policy...?
- *Brown vs. ...* Who?

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What is a Civil Liability Enforcement Mechanism (CLEM)?

- Private citizens enforce through civil lawsuits; “bounty laws”

What is a Civil Liability Enforcement Mechanism (CLEM)?

- Private citizens enforce through civil lawsuits; "bounty laws"
- "Not only unusual, but unprecedented... it is the role of the Supreme Court in our constitutional system that is at stake." (SCOTUS Chief Justice John Roberts)
- Arms race on sensitive issues: **abortion in TX,** guns in CA, more: CLEM Copycats

Criminal vs. Civil Liability

	Criminal	**Civil**
Damages who?	Society	Individual

Criminal vs. Civil Liability

	Criminal	**Civil**
Damages who?	Society	Individual
Burden of Proof?	Beyond a reasonable doubt	A preponderance of evidence (50%)

Criminal vs. Civil Liability

	Criminal	**Civil**
Damages who?	Society	Individual
Burden of Proof?	Beyond a reasonable doubt	A preponderance of evidence (50%)
Rights?	Innocent until proven guilty Right to an attorney No self-incrimination Witnesses in your defense Confront accusing witnesses	None

Case Study: Texas Senate Bill 8: "Heartbeat Bill"

- Took effect in 2021, **during** *Roe v. Wade* era: Timeline of TX Abortion Law
- Bans abortion after detection of fetal heartbeat (6 weeks)
- Civil Liability Enforcement: Minimum $10,000 in damages against any who "aid or abet" an abortion, no fee recovery for defendant
- Very high salience SB-8 is Salient

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- Very high salience SB-8 is Salient
- Strategic inaction $\implies$ avoid creating test cases $\implies$ survives the court system
  - Salient deterrent effect, minimized risk of invalidating SB-8
- Three suits filed against Dr. Alan Braid, dismissed 2.5 yrs later

Economics of CLEM vs. criminal penalties

- Changes the physician optimization problem: increases expected operating costs
  - Can't recover defense costs
  - Malpractice insurance does not cover civil (or criminal) liability
- Lower burden of proof and broader set of affected physicians
  - Criminal penalty targets abortion performers; civil penalty targets any who "aid or abet"
  - Omnipresent threat of scrutiny, hearsay

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People are saying...

People are saying...

HEALTH

1 in 5 Texas OB-GYNs have considered leaving the state over abortions ban, report shows

THE NEW YORKER 100

THE TEXAS OB-GYN EXODUS

Amid increasingly stringent abortion laws, doctors who provide maternal care have been fleeing the state.

By Stephania Taladrid

November 25, 2024

Cost to change state of practice: ≈$150k to $250k

So what do the data say?

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Does the Enforcement Mechansim Matter?

Research Question: **What is the effect of the novel civil liability enforcement in the context of physician migration responses to abortion regulations?**

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Approach

- Triple difference model
    - Texas - all other states
    - Reprod. Health - all other MDs/DOs
    - After SB-8 - Before SB-8
- Novel quarterly panel of universe of US physicians from November 2007 (start of data) through June 2022 (*Dobbs* overturns *Roe v. Wade*)

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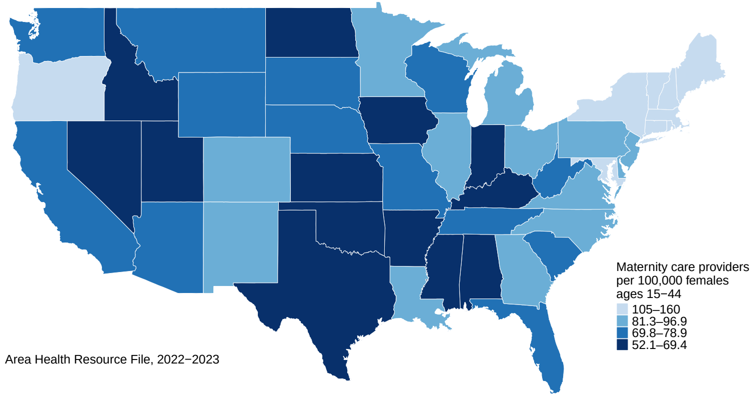
Preview of Results

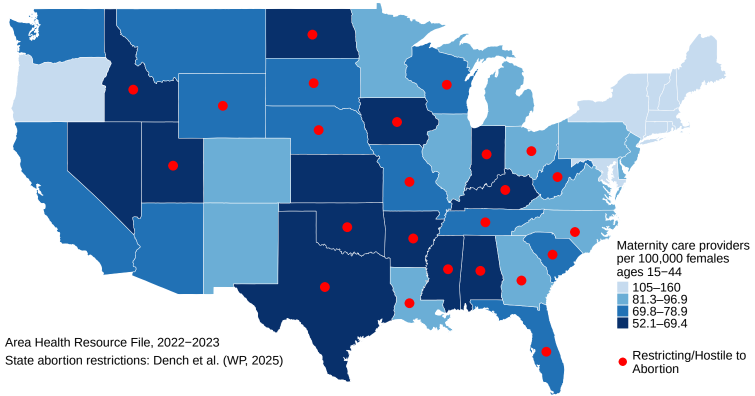
No significant migration response by reproductive health physicians

- Very costly to move states
- Limitation: Retirement margin is unobserved (for now)

We already know: Criminal penalties decrease OB/GYN supply



We already know: Criminal penalties decrease OB/GYN supply



Maternal Mortality

We already know: Criminal penalties decrease OB/GYN supply

- TRAP (Targeted Regulation of Abortion Providers) era, 1993-2022: Timeline of TX Abortion Law
  - 5.3% decrease over 28 years in OB/GYNs in TRAP states Markowski & Vandenbroeck (*Working Paper*, 2025)

We already know: Criminal penalties decrease OB/GYN supply

- TRAP (Targeted Regulation of Abortion Providers) era, 1993-2022: Timeline of TX Abortion Law
  - 5.3% decrease over 28 years in OB/GYNs in TRAP states Markowski & Vandenbroeck (*Working Paper*, 2025)
- Post-*Dobbs* era, 2022-present:
  - 3% decrease over 3 years in OB/GYNs in ban states Diaz-Campo & Pineda-Torres (*Working Paper*, 2025)
  - 0.049% decrease in total population in ban states (52,600/quarter) Dench et al. (*Working Paper*, 2025)

Contribution: But how does civil liability enforcement compare to criminal penalties?

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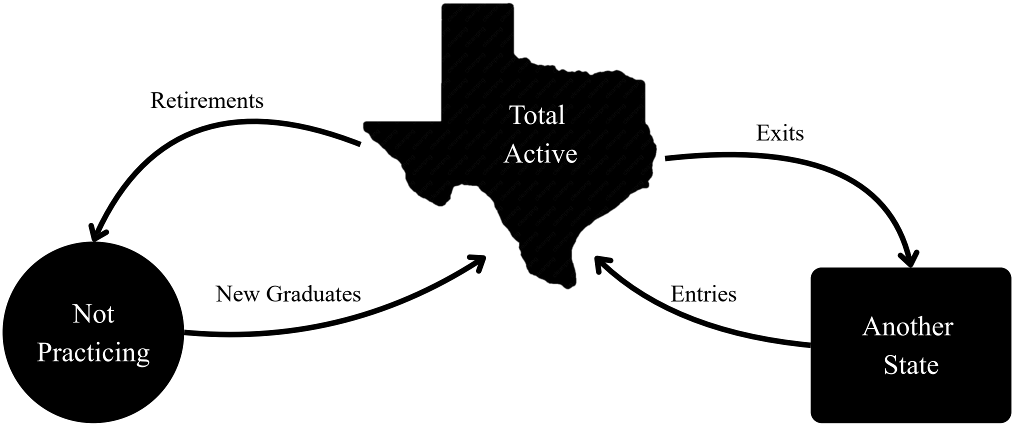
Data

- Novel quarterly panel of the universe of U.S. physicians (CMS NPPES and DAC):
  - Physician name & NPI
  - Practice street address (must be updated within 30 days of moving)
  - Specialty
- Period: Nov. 2007 to June 2022, N ≈ 1.4m physicians
- Migration is defined as:
  - entry into TX, exit out of TX
  - retirement (exiting practice), new graduates (entering practice)

Outcome Variables - Migration



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Summary (by state-specialty-quarter)

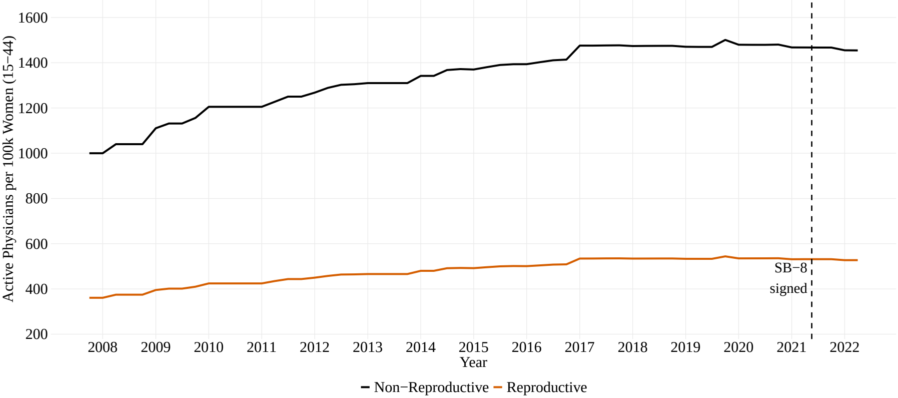
	Mean	SD	Min	Max
Active RHP	161.85	105.27	41	567
Active all other physicians	44.08	92.01	0	1248
Enter state RHP	2.44	5.06	0	92
Exit state RHP	1.62	4.64	0	102
Enter state all physicians	0.87	3.28	0	263
Exit state all physicians	0.60	2.62	0	129

Only OB/GYN

Only ER

Only Family

Trend: Reproductive Providers vs. Other Physicians, all states



OB/GYN

ER

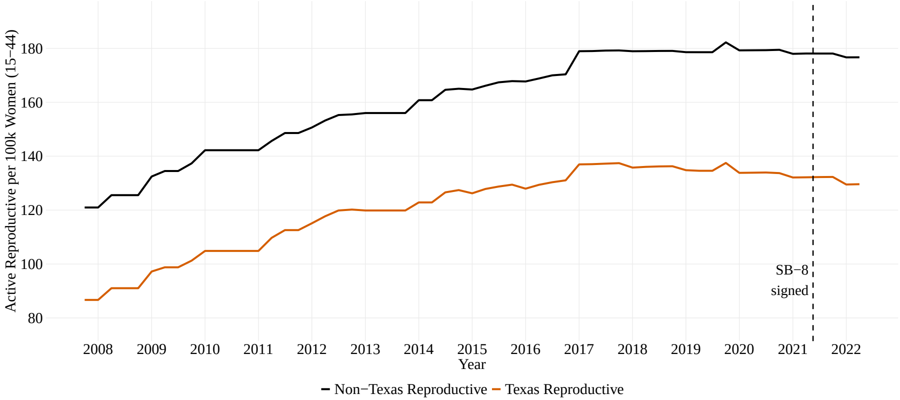
Family

Trend: Reproductive Providers outside vs. inside Texas

OB/GYN

ER

Family

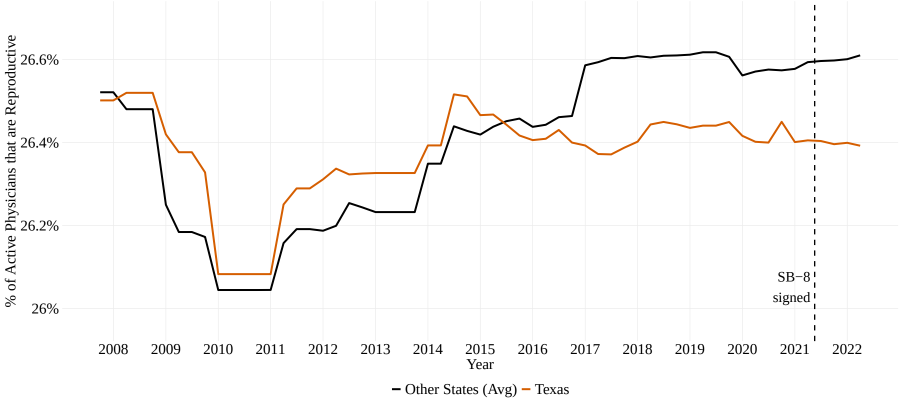


Percentage of Active Physicians: Reproductive Providers

OB/GYN

ER

Family



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Triple Diff: Time, State, & Specialty

Identifying assumption: Had Texas not passed SB-8, RHPs in Texas would behave similarly to RHPs in other states before and after May 19, 2021

$$\text{Outcome}_{ist} = \alpha + \beta_1 \cdot \underbrace{\text{Post SB-8}_t}_{\substack{=1 \text{ if after} \\ \text{May 19, 2021}}} \times \underbrace{\text{Texas}_s}_{\substack{=1 \\ \text{if Texas}}} \times \underbrace{\text{RHP}_i}_{\substack{=1 \\ \text{if Reprod.} \\ \text{Health Phys.}}} + \bar{\beta} \cdot [\text{Post SB-8}_t + \text{Texas}_s + \text{RHP}_i$$

$$+ \text{Post SB-8}_t \times \text{Texas}_s + \text{Post SB-8}_t \times \text{RHP}_i + \text{Texas}_s \times \text{RHP}_i] + \underbrace{u_{ist}}_{\substack{\text{error} \\ \text{term}}}$$

where $\text{Outcome}_{ist} \in \{\text{Total Active, Exit, Entry, Enter Practice, Exit Practice}\}$

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Triple Diff Results

	Active	Exit State	Enter State	Exit Practice	Enter Practice
β_1	−6.33	0.23	0.25	0.01	0.08
	(26.36)	(0.45)	(0.79)	(0.12)	(0.58)
N	100,834	100,834	99,183	100,834	99,183

OLS. SE in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Only OB/GYN Only ER Only Family

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	Active	Exit State	Enter State	Exit Practice	Enter Practice
TX	-12.79^{***}	-0.21^{***}	-0.22^{**}	-0.01	-0.06
	(2.23)	(0.04)	(0.07)	(0.01)	(0.05)
Reproductive	115.32^{***}	0.76^{***}	1.66^{***}	0.07^{***}	0.94^{***}
	(1.08)	(0.02)	(0.03)	(0.005)	(0.02)
Post	3.35^{**}	-0.16^{***}	-0.51^{***}	-0.02^{***}	-0.35^{***}
	(1.10)	(0.02)	(0.03)	(0.005)	(0.02)
TX $\times$ Repro	-26.16^{***}	-0.45^{***}	-0.45	-0.01	-0.08
	(7.68)	(0.13)	(0.23)	(0.03)	(0.17)
TX $\times$ Post	-1.07	0.08	0.13	0.01	0.06
	(7.62)	(0.13)	(0.23)	(0.03)	(0.17)
Repro $\times$ Post	15.10^{***}	-0.38^{***}	-1.22^{***}	-0.07^{***}	-0.93^{***}
	(3.70)	(0.06)	(0.11)	(0.02)	(0.08)
β_1	-6.33	0.23	0.25	0.01	0.08
	(26.36)	(0.45)	(0.79)	(0.12)	(0.58)
Constant	43.75^{***}	0.39^{***}	0.72^{***}	0.02^{***}	0.35^{***}
	(0.32)	(0.01)	(0.01)	(0.001)	(0.01)
N	100,834	100,834	99,183	100,834	99,183

SE in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

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Discussion

Current limitations:

- Low statistical power: one treated state, five treated quarters
- Retirement margin could have largest effects, isn't observable (yet)

Next Steps:

- Synthetic Diff-in-Diff estimation
- Heterogeneity analysis: urban vs. rural counties, physician age
- Border counties: cost of moving practice is lower
- RNs, NPs, PAs may also have a response
- How to observe retirement?

Thank You!

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Summary Statistics: OB/GYN Only

	Mean	SD	Min	Max
Active OB/GYN	92.86	22.96	44	216
Active all other physicians	53.41	100.33	0	1248
Enter state OB/GYN	1.36	2.24	0	44
Exit state OB/GYN	0.92	1.78	0	23
Enter state all physicians	0.87	3.28	0	263
Exit state all physicians	0.60	2.62	0	129

Back

Trend: OB/GYNs vs. Other Physicians

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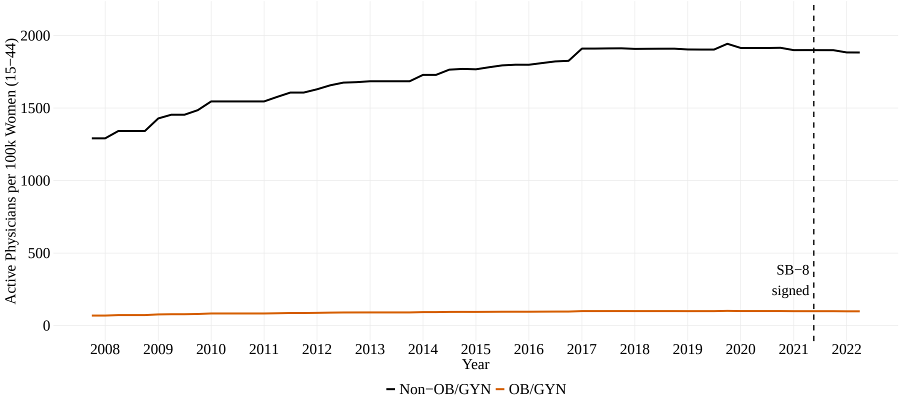

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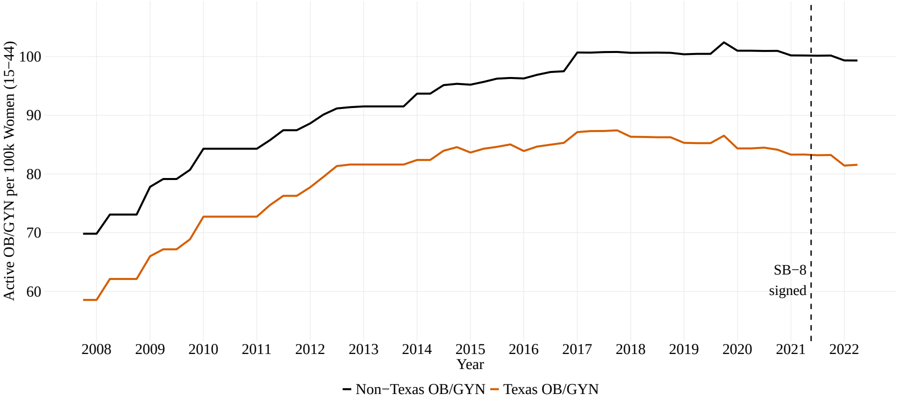
Trend: OB/GYNs outside vs. inside Texas



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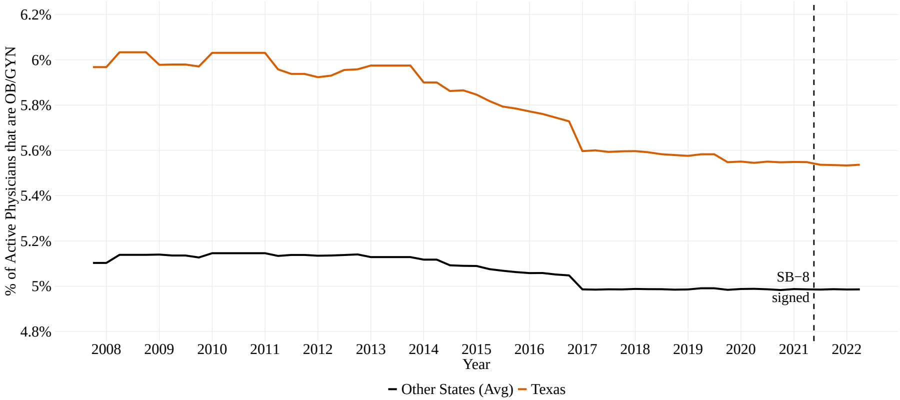
Percentage of Active Physicians: OB/GYN



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Descriptive Means: OB/GYN Only

	Not TX (Pre)	TX (Pre)	Not TX (Post)	TX (Post)
OB/GYN: Active	91.63	79.21	98.20	79.75
OB/GYN: Enter state	1.38	1.08	1.35	0.78
OB/GYN: Exit state	0.73	0.38	1.64	1.01
All other physicians: Active	52.95	37.32	56.35	38.71
All other physicians: Enter state	0.85	0.58	0.89	0.54
All other physicians: Exit state	0.45	0.20	1.09	0.71

Results: OB/GYN Only

	Active	Exit State	Enter State	Exit Practice	Enter Practice
β_1	-3.43	0.06	-0.03	-0.01	-0.11
	(46.98)	(0.76)	(1.35)	(0.20)	(0.98)
N	100,834	100,834	99,183	100,834	99,183

OLS. SE in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

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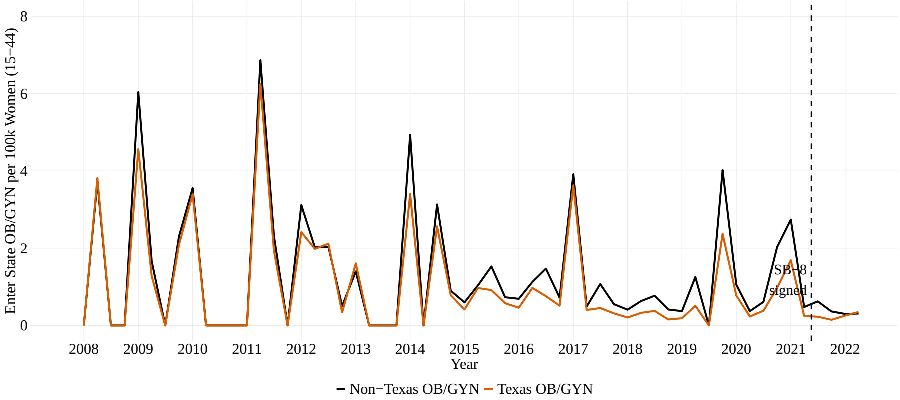
Full Results: OB/GYN Only

	Active	Exit State	Enter State	Exit Practice	Enter Practice
TX	-15.63^{***}	-0.25^{***}	-0.27^{***}	-0.01	-0.08
	(2.30)	(0.04)	(0.07)	(0.01)	(0.05)
OB/GYN	38.68^{***}	0.28^{***}	0.53^{***}	0.02^{*}	0.25^{***}
	(1.92)	(0.03)	(0.06)	(0.01)	(0.04)
Post	4.35^{***}	-0.20^{***}	-0.61^{***}	-0.03^{***}	-0.43^{***}
	(1.13)	(0.02)	(0.03)	(0.005)	(0.02)
TX $\times$ OB/GYN	3.21	-0.11	-0.04	0.01	0.11
	(13.68)	(0.22)	(0.40)	(0.06)	(0.29)
TX $\times$ Post	-1.44	0.10	0.16	0.01	0.08
	(7.84)	(0.13)	(0.22)	(0.03)	(0.16)
OB/GYN $\times$ Post	3.86	-0.11	-0.35	-0.02	-0.25
	(6.58)	(0.11)	(0.19)	(0.03)	(0.14)
β_1	-3.43	0.06	-0.03	-0.01	-0.11
	(46.98)	(0.76)	(1.35)	(0.20)	(0.98)
Constant	52.95^{***}	0.45^{***}	0.85^{***}	0.03^{***}	0.43^{***}

Enter State: OB/GYN



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Exit State: OB/GYN

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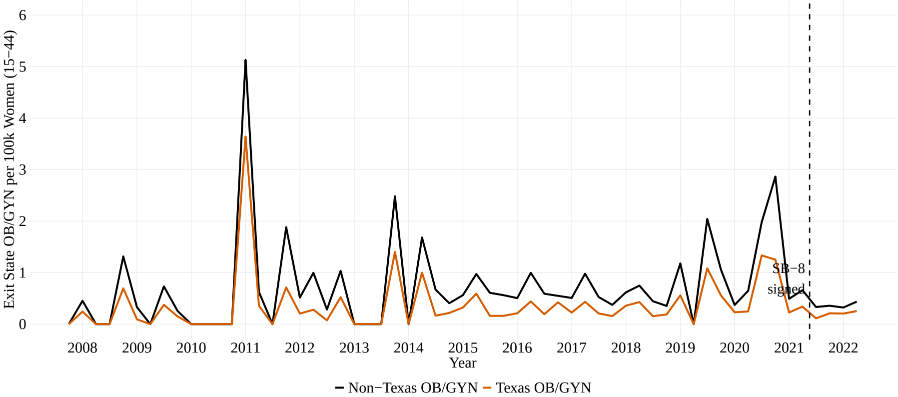

Summary Statistics: ER Only

	Mean	SD	Min	Max
Active ER	103.35	28.24	41	209
Active all other physicians	53.09	100.14	0	1248
Enter state ER	1.63	2.75	0	30
Exit state ER	1.08	2.39	0	33
Enter state all physicians	0.87	3.28	0	263
Exit state all physicians	0.60	2.62	0	129

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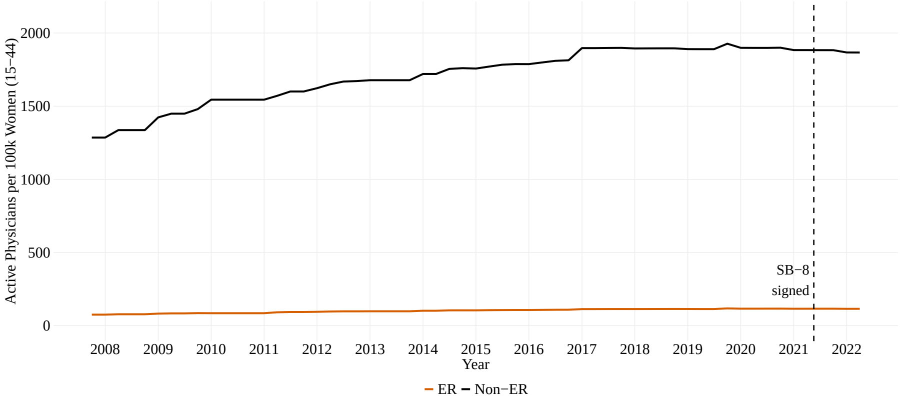
Trend: ER vs. Other Physicians

Back



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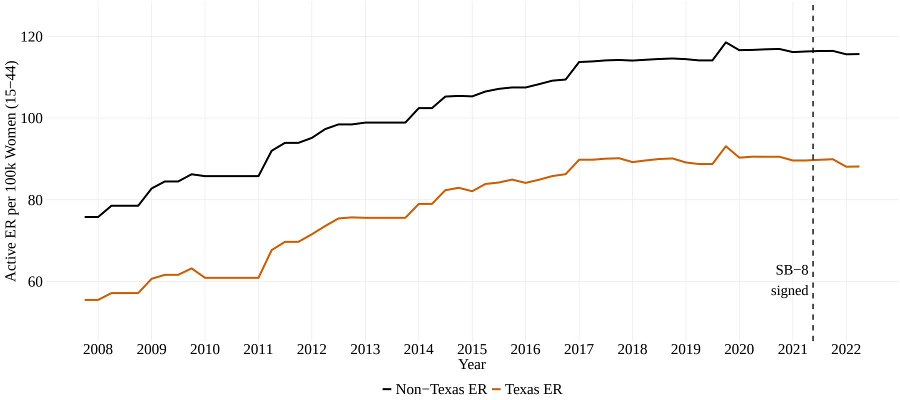
Trend: ER outside vs. inside Texas



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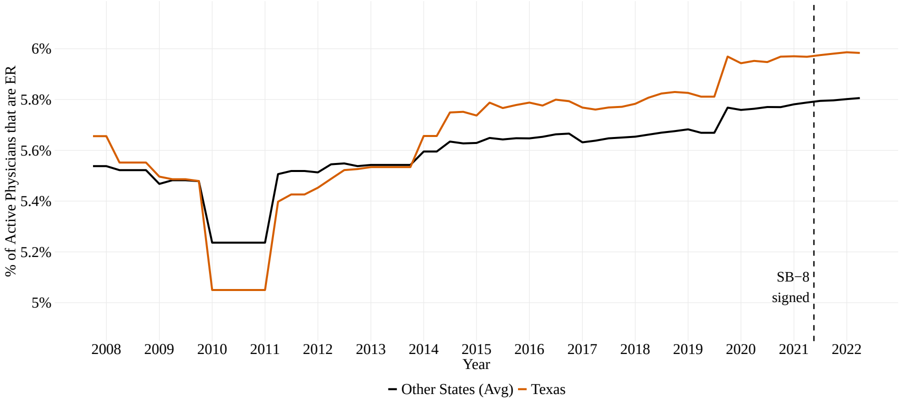
Percentage of Active Physicians: ER



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Descriptive Means: ER Only

	Not TX (Pre)	TX (Pre)	Not TX (Post)	TX (Post)
OB/GYN: Active	91.63	79.21	98.20	79.75
OB/GYN: Enter state	1.38	1.08	1.35	0.78
OB/GYN: Exit state	0.73	0.38	1.64	1.01
All other physicians: Active	52.95	37.32	56.35	38.71
All other physicians: Enter state	0.85	0.58	0.89	0.54
All other physicians: Exit state	0.45	0.20	1.09	0.71

Results: ER Only

	Active	Exit State	Enter State	Exit Practice	Enter Practice
β_1	−1.71	0.10	0.04	−0.001	0.02
	(46.92)	(0.76)	(1.35)	(0.20)	(0.98)
N	100,834	100,834	99,183	100,834	99,183

OLS. SE in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

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Full Results: ER Only

	Active	Exit State	Enter State	Exit Practice	Enter Practice
TX	-15.28^{***}	-0.24^{***}	-0.26^{***}	-0.01	-0.07
	(2.29)	(0.04)	(0.07)	(0.01)	(0.05)
ER	48.31^{***}	0.32^{***}	0.77^{***}	0.01	0.38^{***}
	(1.91)	(0.03)	(0.06)	(0.01)	(0.04)
Post	4.15^{***}	-0.19^{***}	-0.61^{***}	-0.03^{***}	-0.42^{***}
	(1.13)	(0.02)	(0.03)	(0.005)	(0.02)
TX $\times$ ER	-8.48	-0.18	-0.13	0.001	-0.02
	(13.66)	(0.22)	(0.40)	(0.06)	(0.29)
TX $\times$ Post	-1.49	0.10	0.16	0.01	0.07
	(7.83)	(0.13)	(0.22)	(0.03)	(0.16)
ER $\times$ Post	10.95	-0.16	-0.51^{**}	-0.01	-0.38^{**}
	(6.57)	(0.11)	(0.19)	(0.03)	(0.14)
β_1	-1.71	0.10	0.04	-0.001	0.02
	(46.92)	(0.76)	(1.35)	(0.20)	(0.98)
Constant	52.66^{***}	0.45^{***}	0.84^{***}	0.03^{***}	0.42^{***}

Enter State: ER

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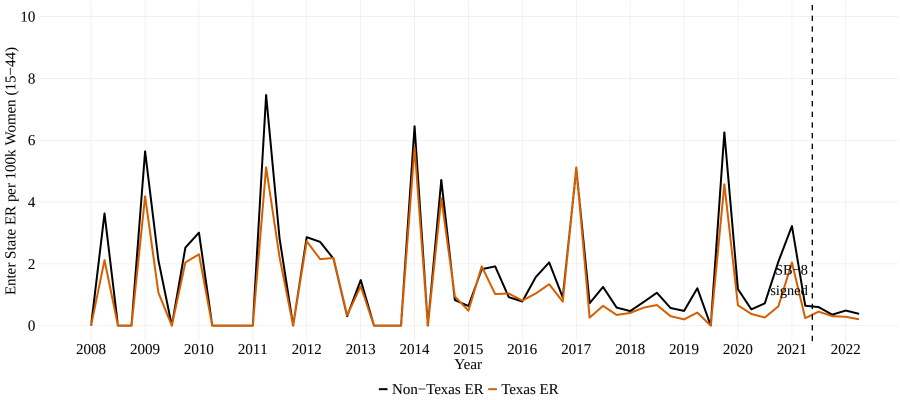
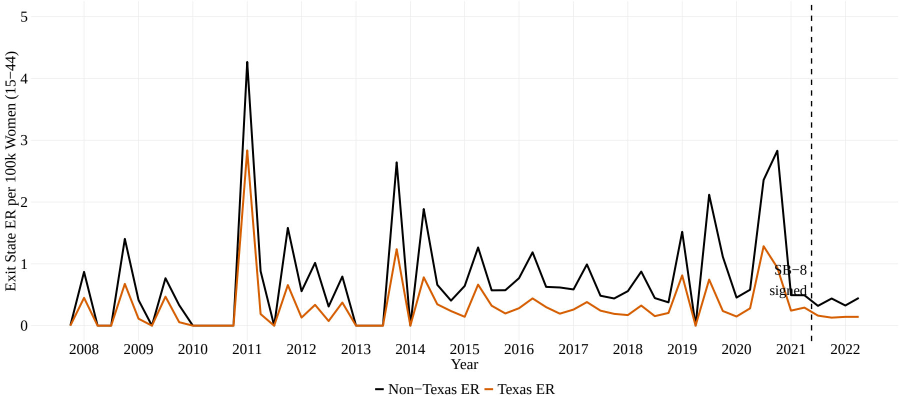


Exit State: ER



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Summary Statistics: Family Medicine Only

	Mean	SD	Min	Max
Active Family	289.35	86.50	97	567
Active all other physicians	47.39	90.32	0	1248
Enter state Family	4.33	7.67	0	92
Exit state Family	2.87	7.31	0	102
Enter state all physicians	0.87	3.28	0	263
Exit state all physicians	0.60	2.62	0	129

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Trend: Family Medicine vs. Other Physicians

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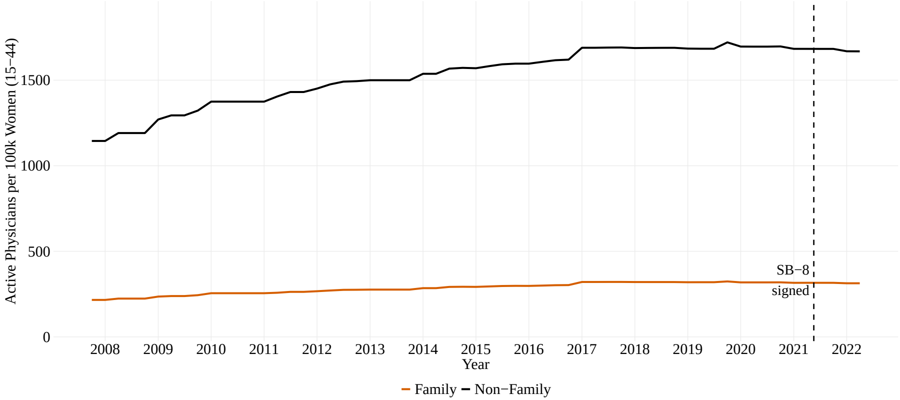
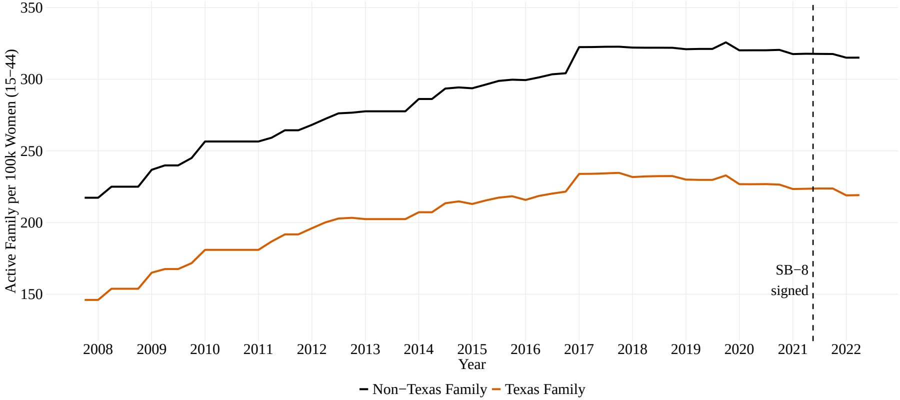


Trend: Family Medicine outside vs. inside Texas



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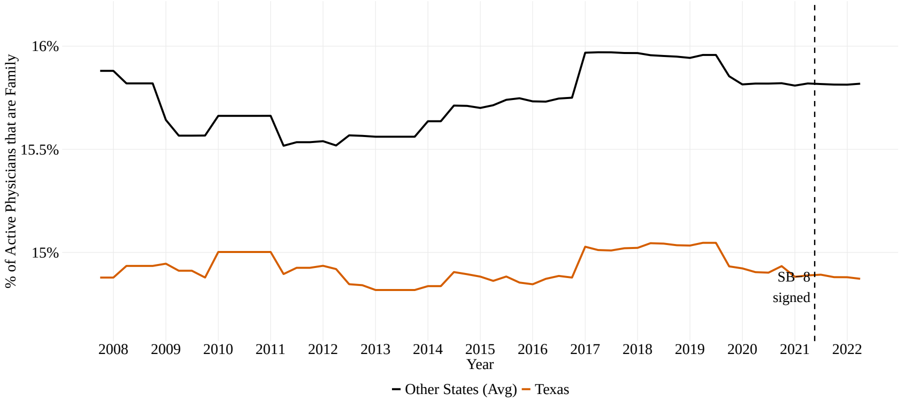
Percentage of Active Physicians: Family Medicine



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Descriptive Means: Family Medicine Only

	Not TX (Pre)	TX (Pre)	Not TX (Post)	TX (Post)
OB/GYN: Active	91.63	79.21	98.20	79.75
OB/GYN: Enter state	1.38	1.08	1.35	0.78
OB/GYN: Exit state	0.73	0.38	1.64	1.01
All other physicians: Active	52.95	37.32	56.35	38.71
All other physicians: Enter state	0.85	0.58	0.89	0.54
All other physicians: Exit state	0.45	0.20	1.09	0.71

Results: Family Medicine Only

	Active	Exit State	Enter State	Exit Practice	Enter Practice
β_1	−12.92	0.50	0.67	0.03	0.31
	(42.87)	(0.75)	(1.32)	(0.20)	(0.97)
N	100,834	100,834	99,183	100,834	99,183

OLS. SE in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

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Full Results: Family Medicine Only

	Active	Exit State	Enter State	Exit Practice	Enter Practice
TX	-13.29^{***}	-0.22^{***}	-0.23^{***}	-0.01	-0.06
	(2.10)	(0.04)	(0.07)	(0.01)	(0.05)
Family	237.63^{***}	1.55^{***}	3.36^{***}	0.16^{***}	2.01^{***}
	(1.75)	(0.03)	(0.05)	(0.01)	(0.04)
Post	3.77^{***}	-0.18^{***}	-0.54^{***}	-0.02^{***}	-0.37^{***}
	(1.03)	(0.02)	(0.03)	(0.005)	(0.02)
TX $\times$ Family	-67.39^{***}	-0.98^{***}	-1.07^{**}	-0.03	-0.30
	(12.48)	(0.22)	(0.39)	(0.06)	(0.28)
TX $\times$ Post	-1.23	0.09	0.14	0.01	0.06
	(7.15)	(0.13)	(0.22)	(0.03)	(0.16)
Family $\times$ Post	28.27^{***}	-0.78^{***}	-2.57^{***}	-0.16^{***}	-2.01^{***}
	(6.01)	(0.11)	(0.19)	(0.03)	(0.14)
β_1	-12.92	0.50	0.67	0.03	0.31
	(42.87)	(0.75)	(1.32)	(0.20)	(0.97)
Constant	47.00^{***}	0.41^{***}	0.77^{***}	0.02^{***}	0.37^{***}

Enter State: Family Medicine

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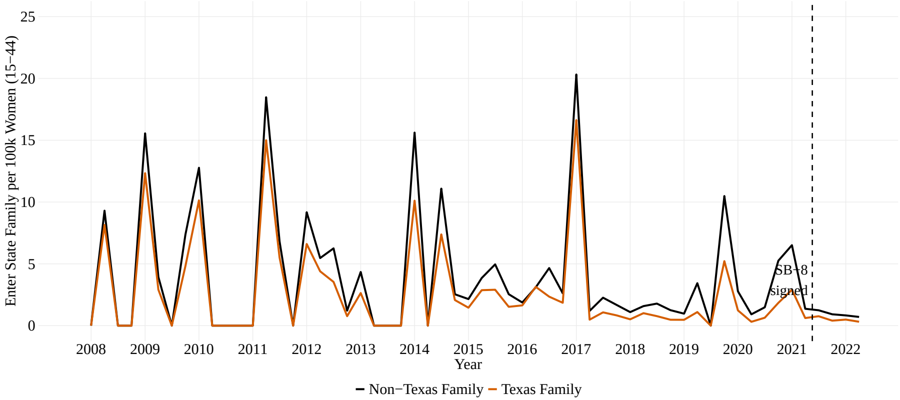
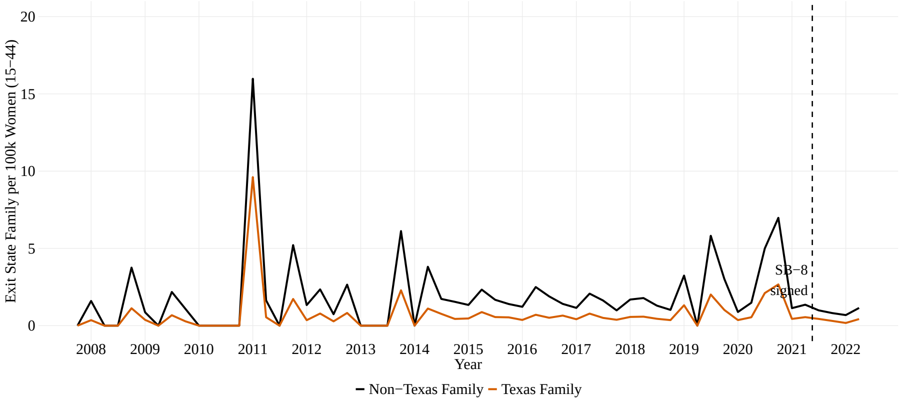


Exit State: Family Medicine



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Enter State: Reproductive Providers

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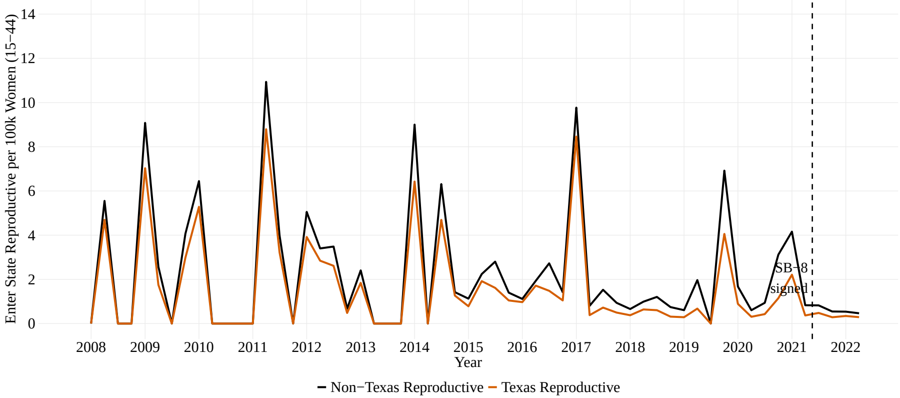

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Exit State: Reproductive Providers

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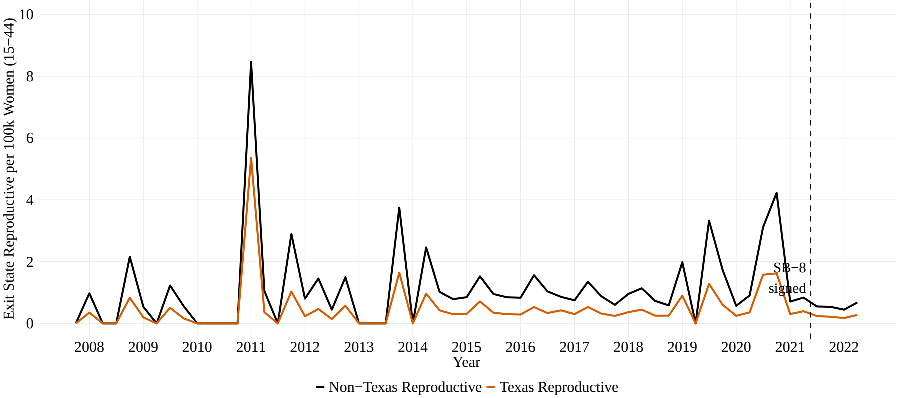

Timeline of Recent Abortion Law in Texas

SEP. 1, 2021
Senate Bill 8 takes effect in Texas.

SEP. 20, 2021
First lawsuit filed against Texan abortion provider.

JUN. 24, 2022
Dobbs vs. JWHO overturns *Roe v. Wade*.

JUL. 24, 2022
Texas Trigger Law takes effect, fully banning abortion.

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CLEM Copycats

State	Subject	Status	Bill Number
Texas	Abortion pill providers	Enacted	HB 7
Texas	Transgender bathrooms	Enacted	SB 8 (Different, 89th Leg.)
Idaho	Library censorship	Enacted	HB 710
California	Social media hate speech	Pending	SB 771
California	Gun control	Enacted	SB 1327
Texas	Gun control	Failed	HB 925
Illinois	Gun control	Failed	HB 4156
New York	Gun control	Announced	N/A

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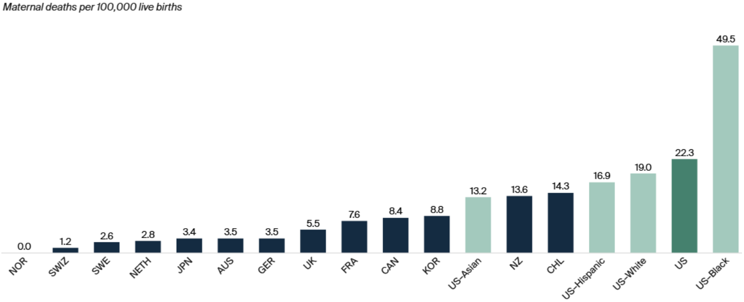
The United States continues to have the highest maternal death rate, with the rate for Black women by far the highest of any group.



Notes: The maternal mortality ratio is defined as the death of a woman while pregnant or within 42 days of termination of pregnancy, irrespective of the duration and site of the pregnancy from any cause related to or aggravated by the pregnancy or its management but not from accidental or incidental causes. For more information on how maternal mortality is defined, see Organisation for Economic Co-operation and Development, "Maternal and Infant Mortality," in *Health at a Glance 2023: OECD Indicators* (OECD, 2023). 2015 data for FRA; 2017 data for UK; 2018 data for NZ; 2020 data for CAN and SWIZ; 2021 data for AUS, GER, JPN, KOR, NETH, and SWE; 2022 data for CHL (provisional), NOR, and US. Due to sample size limitations, data for US–AIAN cannot be displayed. AIAN = American Indian and Alaska Native. Asian Americans include a wide range of distinct communities. Such groupings are imperfect, as they mask significant difference in maternal mortality rates.

Data: All country data from OECD Health Statistics 2023 extracted on February 29, 2024, except data for US are 2022 data from the Centers for Disease Control and Prevention, Nationa Center for Health Statistics, National Vital Statistics System, mortality and natality data files, "Maternal Mortality Rates in the United States, 2022."

Source: Munira Z. Gunja et al., *Insights into the U.S. Maternal Mortality Crisis: An International Comparison* (Commonwealth Fund, June 2024). https://doi.org/10.26099/cthn-st75

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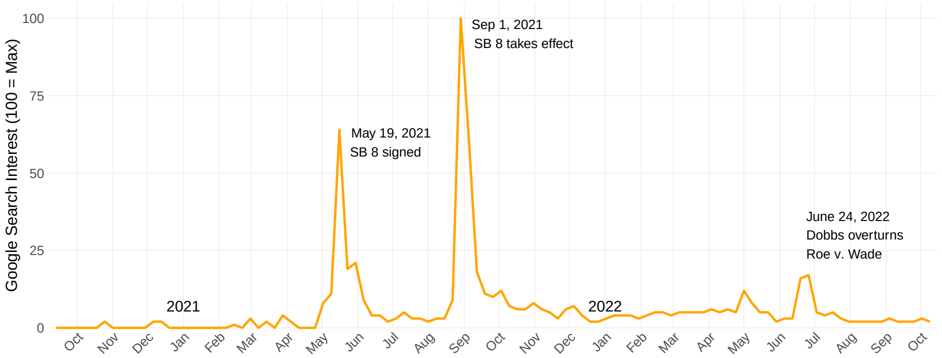
Google searches for "Texas Heartbeat" suggest salience



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